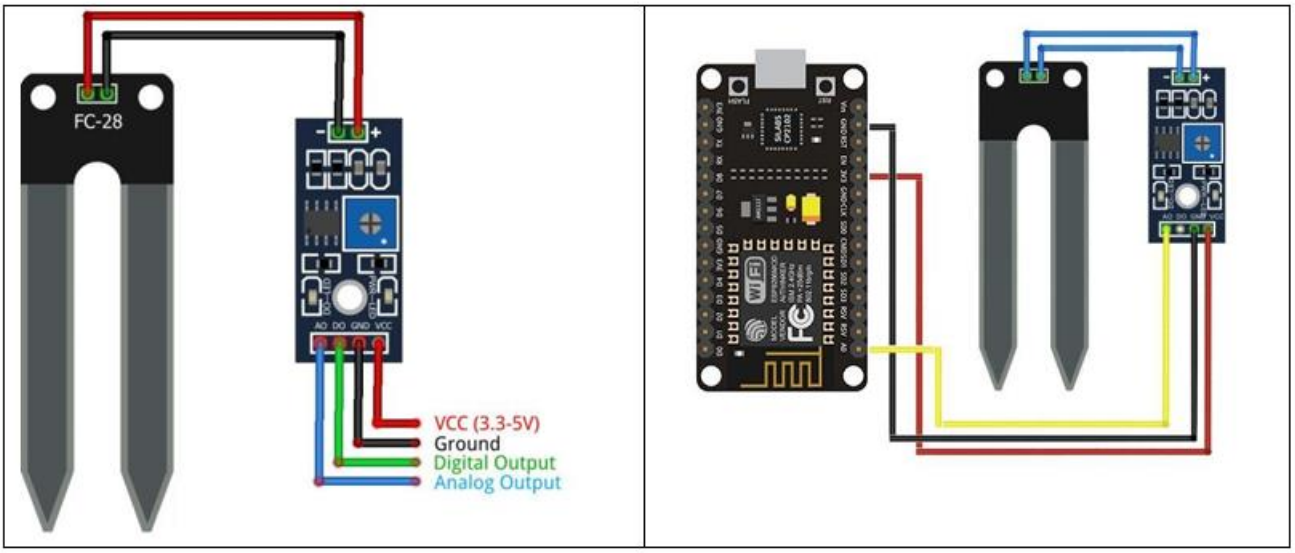
ASSIGNMENT – 7

BATCH 8

Aim: To measure the soil moisture level using a soil moisture sensor and display the moisture percentage on the Blynk IoT Cloud through Wi-Fi and without Blynk as well. The system can also notify the user when the soil becomes dry.

Working Principle

The soil moisture sensor measures the electrical conductivity of the soil.
• Wet soil conducts electricity better.
• Dry soil has higher resistance.
The sensor outputs an analog voltage proportional to the moisture content.
The ESP8266:
1. Reads analog value from sensor.
2. Converts it into moisture percentage.
3. Sends data to Blynk Cloud.
4. Blynk App displays moisture in real time.
5. If moisture is below threshold, a notification or LED can be activated.



Without Blynk

Code:

```
#define SOIL_PIN A0

void setup() {
  Serial.begin(9600);
}

void loop() {
  int moistureValue = analogRead(SOIL_PIN);

  Serial.print("Moisture: ");
  Serial.println(moistureValue);

  if (moistureValue > 700) {
    Serial.println("DRY");
  }
  else if (moistureValue > 400) {
    Serial.println("MOIST");
  }
  else {
    Serial.println("WET");
  }

  delay(1000);
}
```

Output:



Without Blynk

Code:

```
#define BLYNK_TEMPLATE_ID "TMPL3Z-p8Xddg"
#define BLYNK_TEMPLATE_NAME "Soil Moisture"
#define BLYNK_AUTH_TOKEN "As4Ak4Z77IXhImChofsbvWyhP50Y4zz9"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

char ssid[]="Not for Sale";
char pass[]="panipuri";

BlynkTimer timer;

#define SOIL_PIN A0

void sendSoilMoisture()
{
int s=analogRead(SOIL_PIN);
int m=map(s,720,280,0,100);
m=constrain(m,0,100);

Serial.print("Raw: ");
Serial.print(s);
Serial.print("  Moisture: ");
Serial.print(m);
Serial.println("%");

Blynk.virtualWrite(V0,m);
}

void setup()
{
Serial.begin(115200);
delay(1000);

Serial.println();
Serial.println("Connecting WiFi...");

WiFi.mode(WIFI_STA);
WiFi.begin(ssid,pass);

while(WiFi.status()!=WL_CONNECTED)
{
delay(500);
Serial.print(".");
}

Serial.println();
Serial.println("WIFI OK");
Serial.print("IP: ");
Serial.println(WiFi.localIP());

Blynk.config(BLYNK_AUTH_TOKEN);
```

```
Serial.println("Connecting to Blynk...");

if(Blynk.connect(10000))
Serial.println("BLYNK CONNECTED");
else
Serial.println("BLYNK FAILED");

timer.setInterval(2000L,sendSoilMoisture);
}

void loop()
{
Blynk.run();
timer.run();
}
```

Output:

